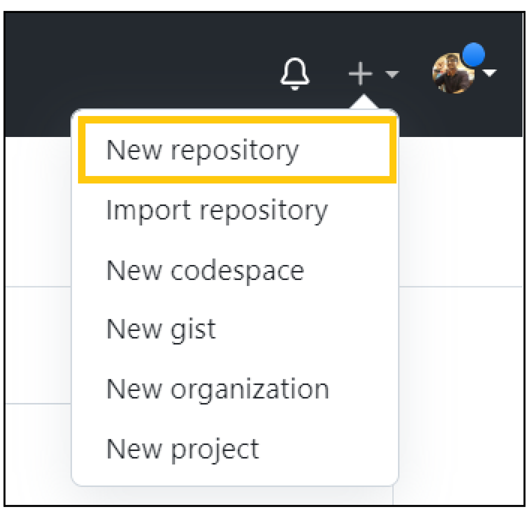
Stock Prices Insights Generation Application

Procedure in a nutshell

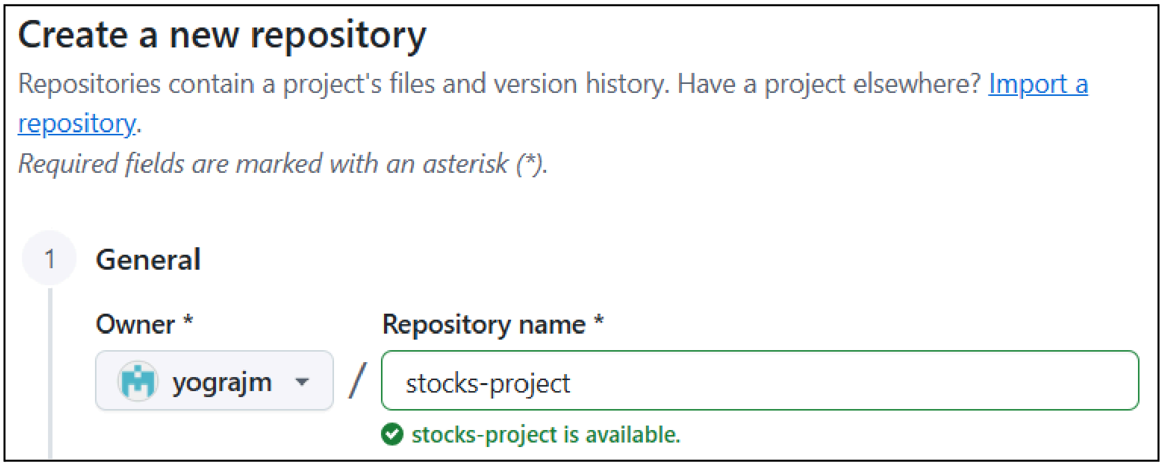
1. Download the Application folder zip file and unzip it
2. Create a new GitHub repository
3. Upload the content of Application folder to this repository
4. Add OpenAI API key into Codespaces Secrets of this repository
5. Create a new Codespace
6. Create and activate virtual environment
7. Install requirements
8. Run main.py file to load data from pickle files and save it into a SQLite DB
9. Run app.py to launch the application
10. Once done, close the application and delete the codespace

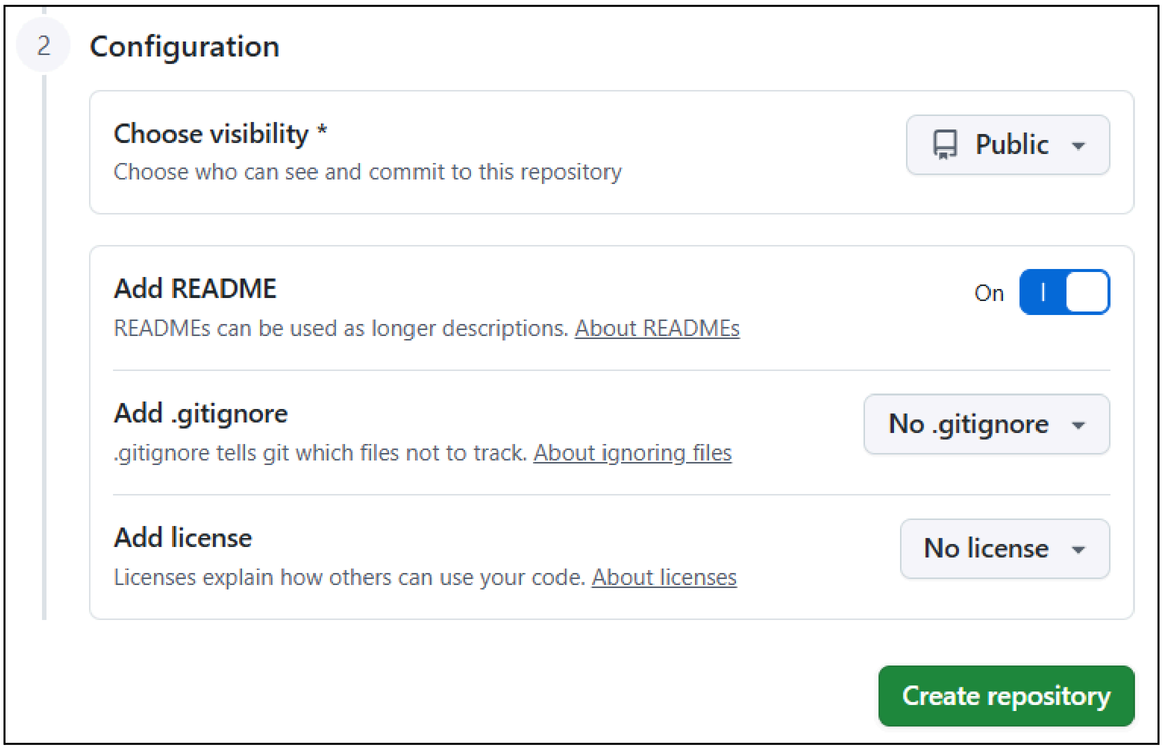
Steps

1. First, create an account on GitHub if you don't have one already.
2. Once the account is created, create a new repository

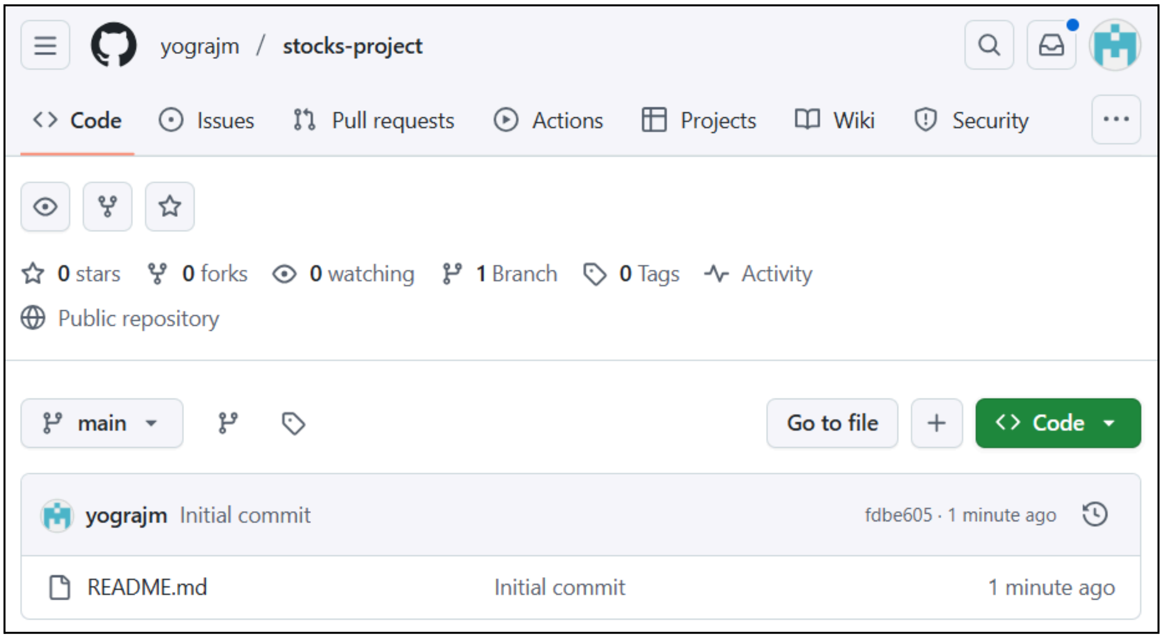


3. Fill in the details for your repository. Give the ***repository name***, turn on the ***Add README*** button, and select ***Create repository***.

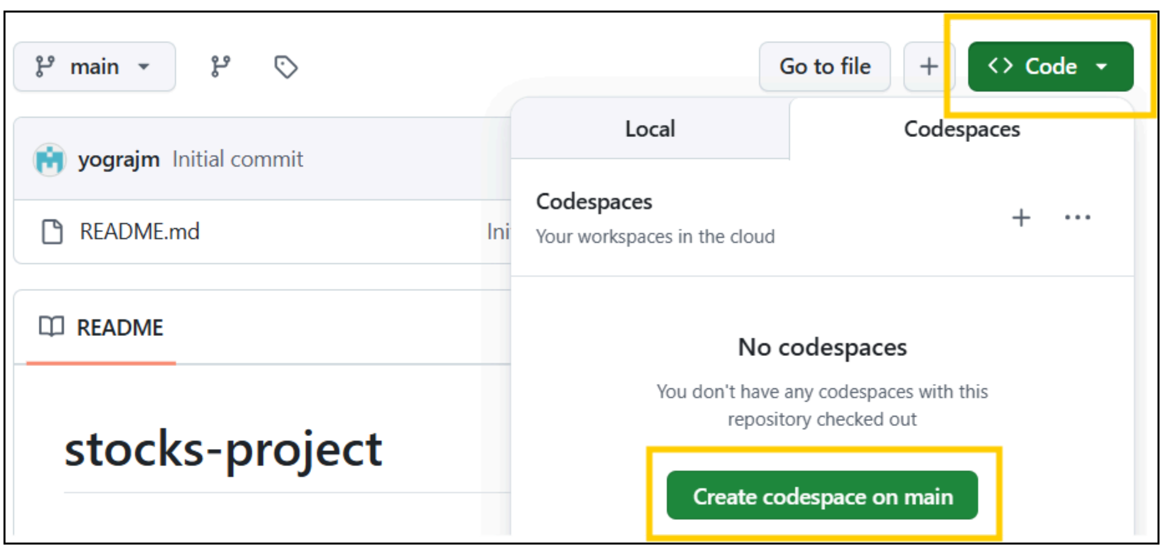

Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1
General
Owner *
yograjm
/
Repository name *
stocks-project
stocks-project is available.


2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
On
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
Licenses explain how others can use your code. About licenses
No license
Create repository

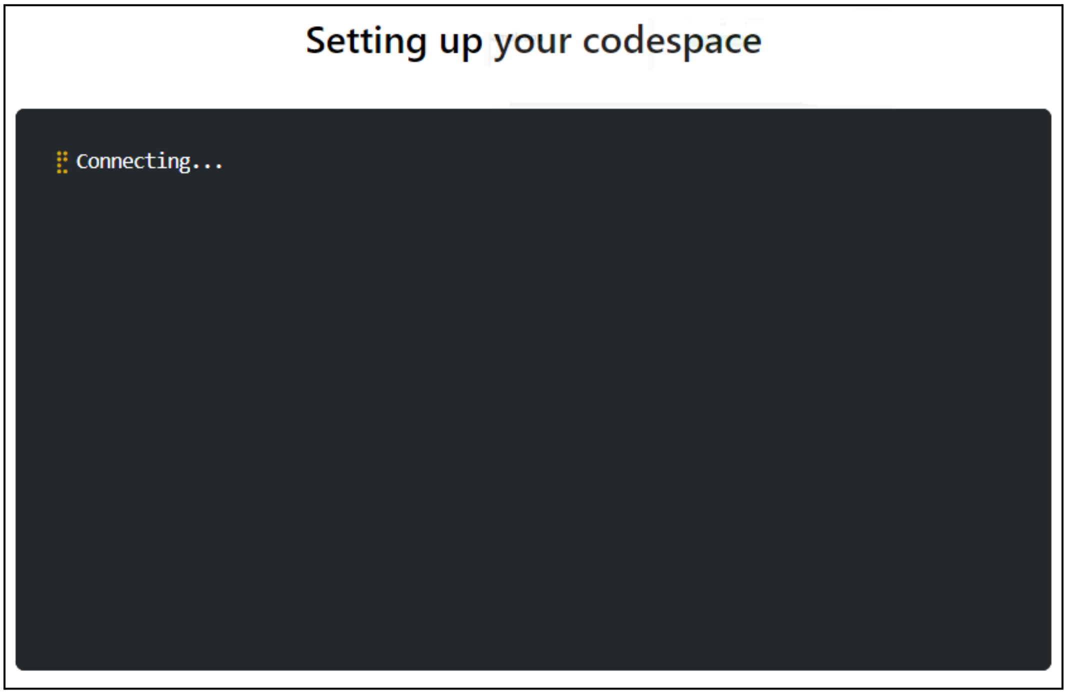
4. Your repository should be created as shown below.



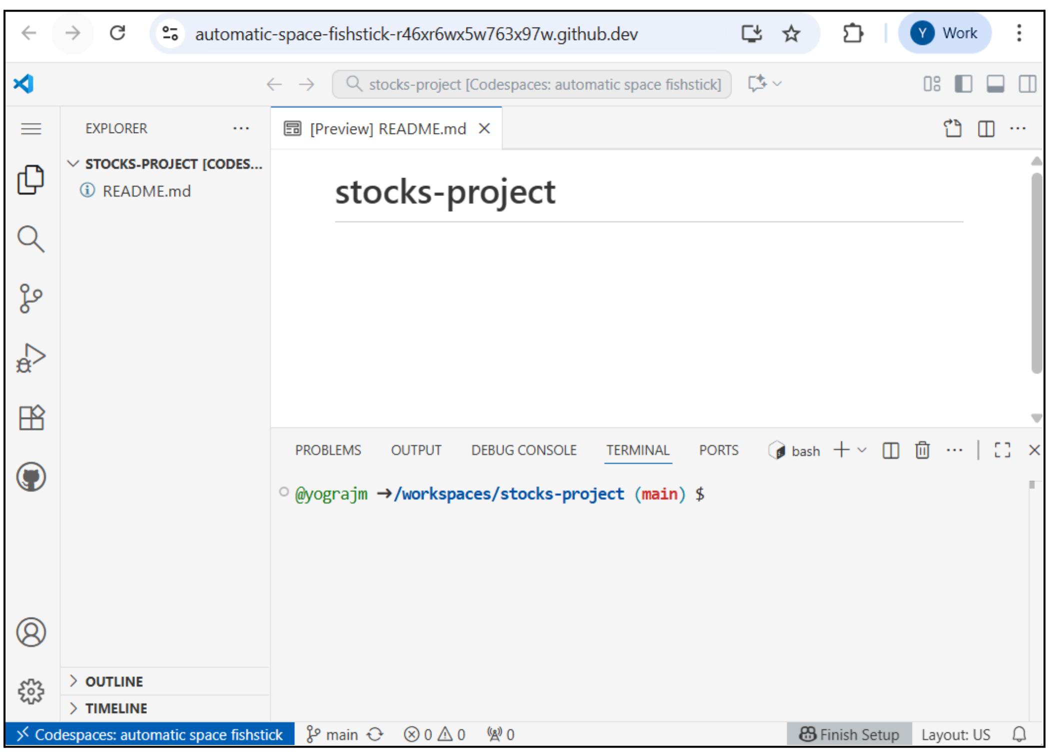
5. Now, start a new Codespace by going to the '***Code***' dropdown. Select the Codespaces tab, and click on '**Create codespace on main**'.



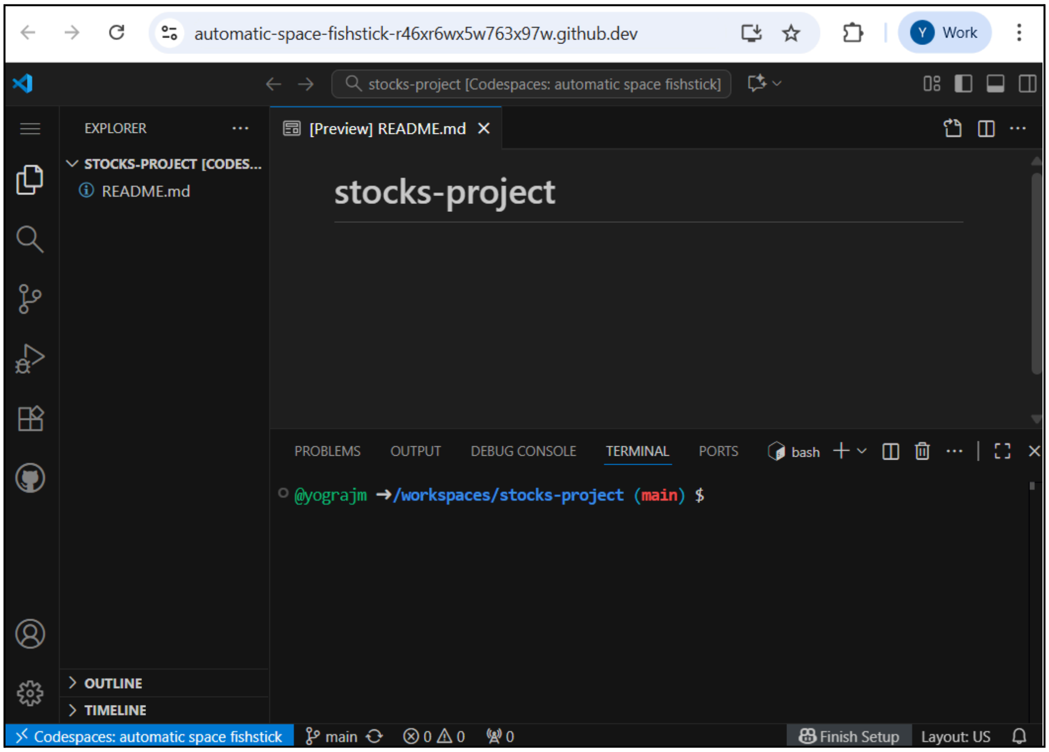
6. It will open a new tab, and start *Setting up your codespace*.



7. After loading, the Editor should open in the browser similar to below.

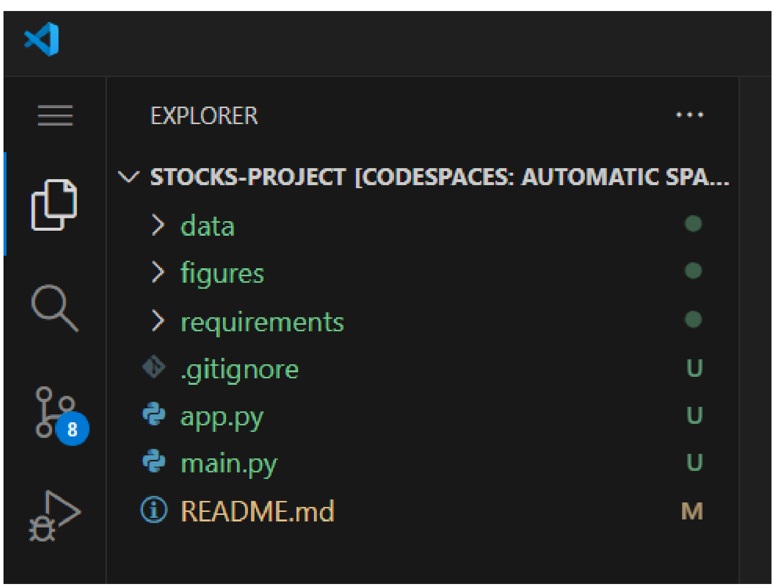


8. By default it is Light mode. You can change the theme by going to
Files -> Preferences -> Theme -> Color theme -> Default Dark Modern.

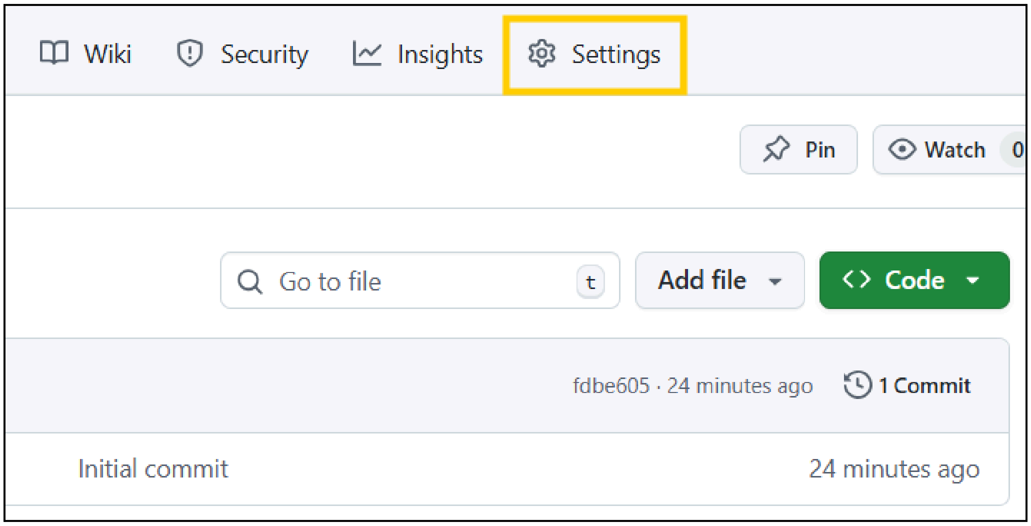


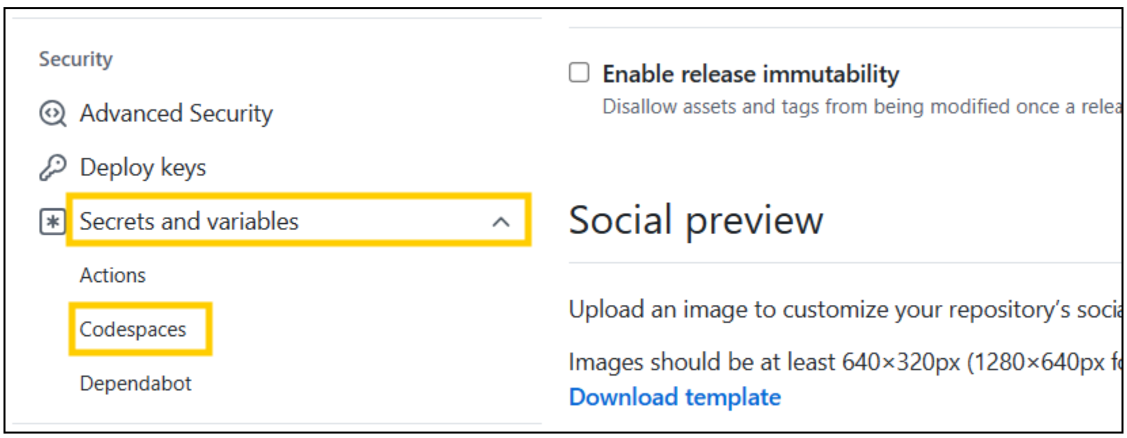
9. Download the [***Application***]() folder zip file and unzip it in your system. Upload the files from the Application folder to your GitHub Codespace by drag-and-drop. Replace the existing README file with the one present inside the Application folder.

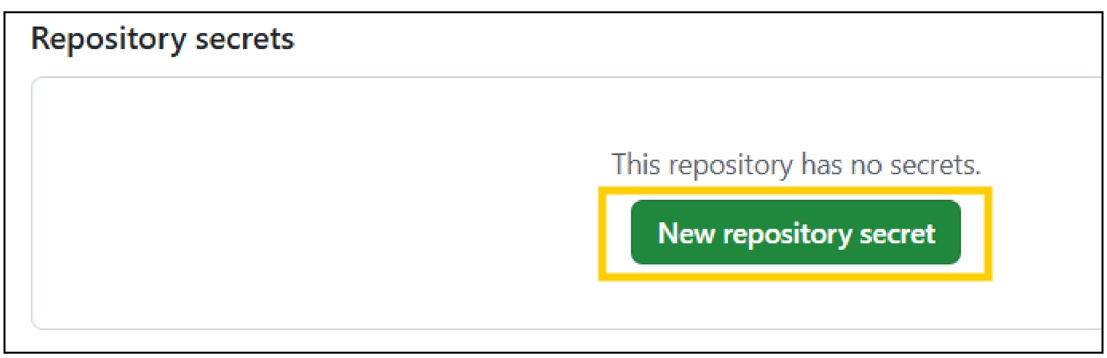
10. Application files should be visible on the GitHub Codespace.

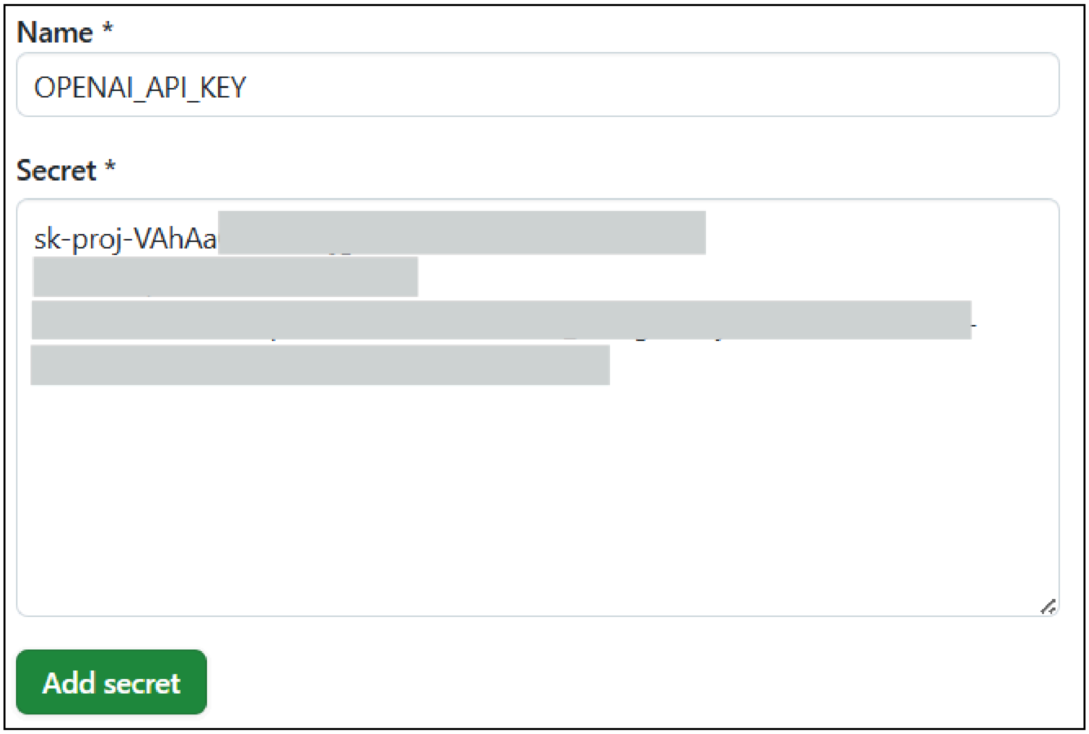


11. Next, Add your OpenAI API key to your GitHub repository ***Secrets***.
    Go to ***Settings -> Secrets and Variables -> Codespaces -> New repository secret ->*** Give Name (eg. **OPENAI_API_KEY**) and paste Secret Value (API key value) ***-> Add secret***



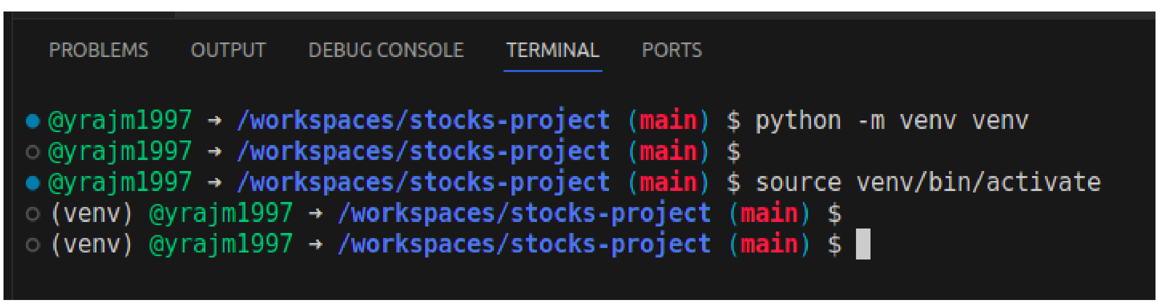






12. Once the Secret is updated, reload your GitHub Codespace to reflect the changes.
13. Next, create and activate a virtual environment by running the below commands:

```
python -m venv venv
source venv/bin/activate
```



14. Install requirements *(This step may take some time to execute)*

```
pip install -r requirements/requirements.txt
```

```
○ (venv) @yrajm1997 → /workspaces/stocks-project (main) $ pip install -r requirements/requirements.txt
  Collecting numpy (from -r requirements/requirements.txt (line 1))
    Obtaining dependency information for numpy from https://files.pythonhosted.org/packages/9b/b4/e3c7e6
  p312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
    Downloading numpy-2.1.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (60 kB)
       ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.9/60.9 kB 723.8 kB/s eta 0:00:00
  Collecting pandas (from -r requirements/requirements.txt (line 2))
```

15. Run the main.py file to load the data from pickle files and save it inside a SQLite DB.

```
python main.py
```

```
● (venv) @yrajm1997 → /workspaces/stocks-project (main) $ python main.py
  File 'stock_db.sqlite' does not exist. Creating new...
  Created DB successfully!

  Table 'stock_prices' created successfully!
  Data inserted into 'stock_prices' successfully!

  Table 'income_statement' created successfully
  Data inserted into 'income_statement' successfully!

  Table 'balancesheet_statement' created successfully
  Data inserted into 'balancesheet_statement' successfully!

  Table 'cashflow_statement' created successfully
  Data inserted into 'cashflow_statement' successfully!
  DB connection closed!
```

After this step, you should have the 'stock_db.sqlite' file available within the codespace.

16. To start the application run the following command:

```
chainlit run app.py
```

```
○ (venv) @yrajm1997 → /workspaces/stocks-project (main) $ chainlit run app.py
  2024-10-25 12:15:52 - Created default config file at /workspaces/stocks-project/
  2024-10-25 12:15:52 - Created default translation directory at /workspaces/stock
  2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
  2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
  2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
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  2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
  2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
  2024-10-25 12:15:55 - Python REPL can execute arbitrary code. Use with caution.
  2024-10-25 12:15:55 - Created default chainlit markdown file at /workspaces/stoc
  2024-10-25 12:15:55 - Your app is available at http://localhost:8000
```
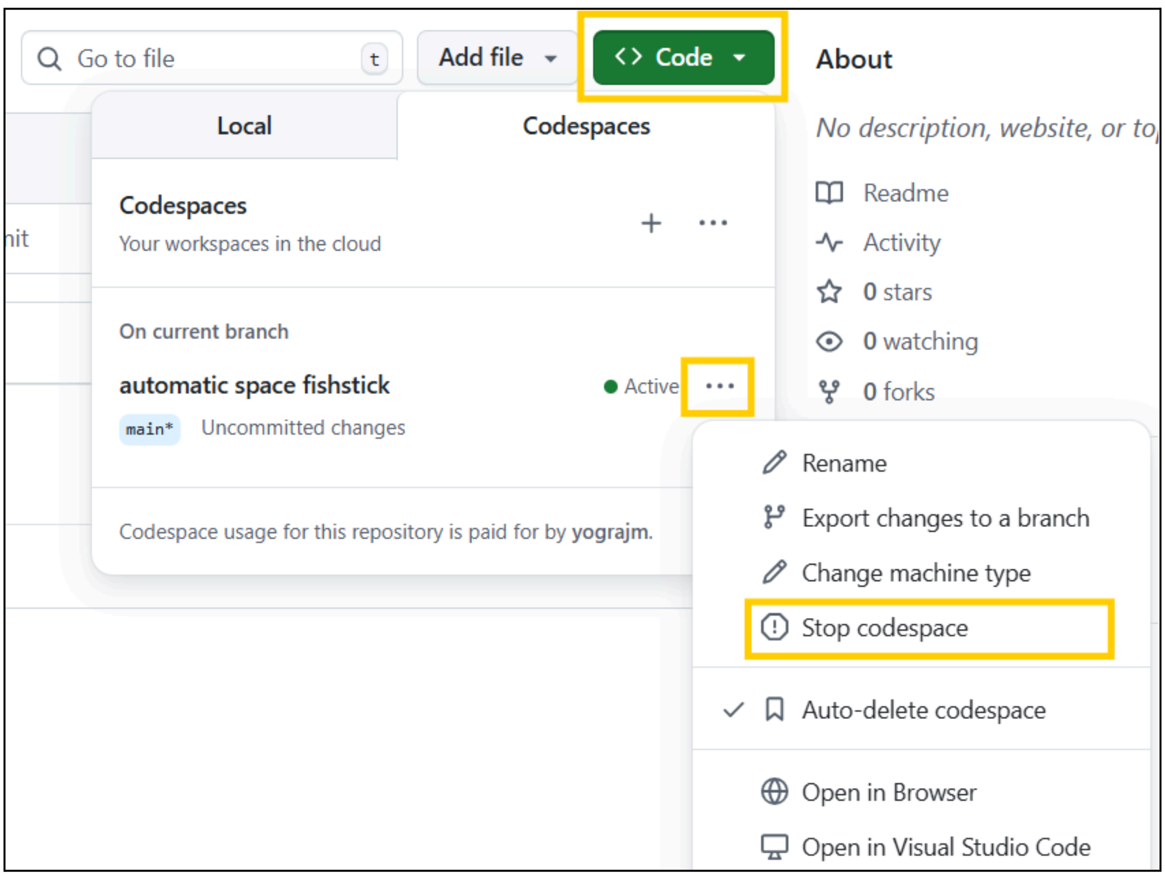
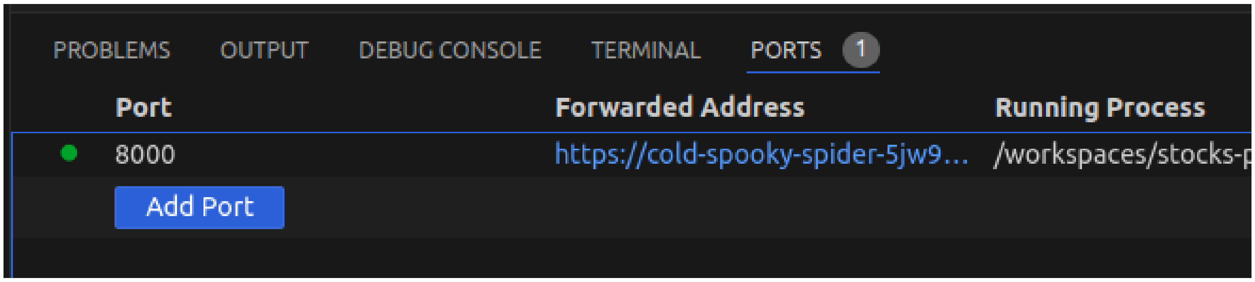
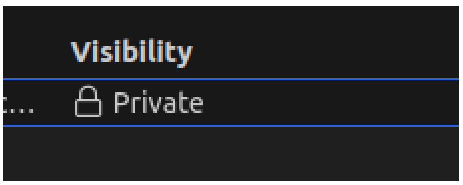

If it shows an error like `KeyError: 'OPENAI_API_KEY'`, try stopping your GitHub Codespace and starting it again.



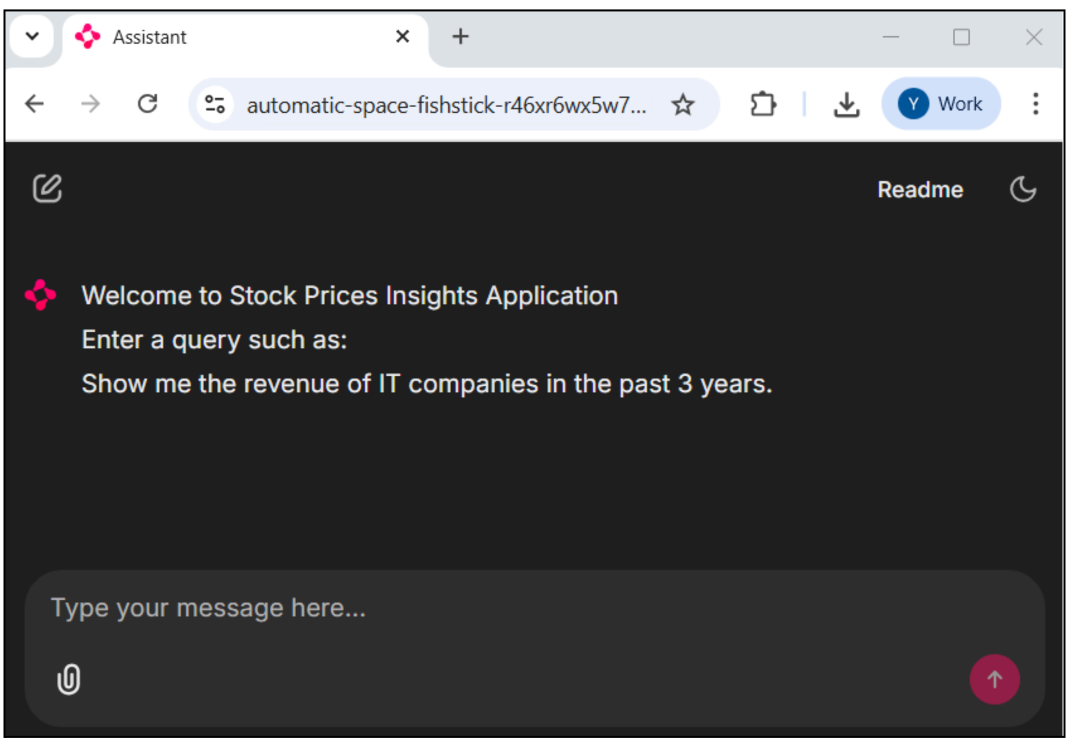
17. The app should be accessible at port 8000 of localhost. You have to forward the port to access it in the browser.
Go to the **PORTS** tab beside the **TERMINAL** tab. The port should already be forwarded, else you can click on 'Add Port' and provide the port number.



Similarly, you can make the port visibility to 'public' to share it with others.
Do right click on it, then ***Port Visibility -> Public***.



18. Once the port is forwarded, you can access its 'Forwarded Address' to access the application.



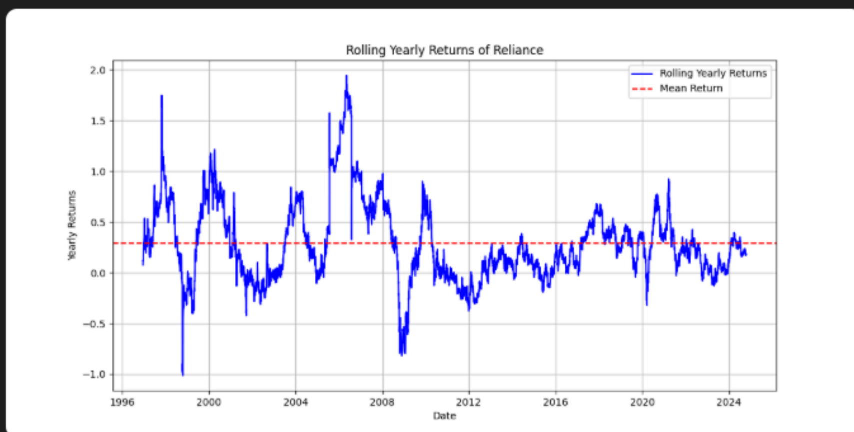
19. Try with some user inputs:

Prompt: *I want to understand the distribution of yearly returns of Reliance. Can you help me plot the rolling yearly returns highlighting the mean returns using a mean return line.*

I want to understand the distribution of yearly returns of Reliance. Can you help me plot the rolling yearly returns highlighting the mean returns using a mean return line.

Response:

Mean Yearly Return: 0.2889



Further suggestions:

You can ask for the historical stock prices of Reliance to calculate the yearly returns. Additionally, request the average yearly return to plot the mean return line. Consider asking for the specific date range you want to analyze for more accurate insights.

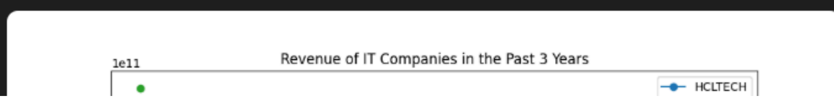
Prompt: *Show me the revenue of IT companies in the past 3 years.*

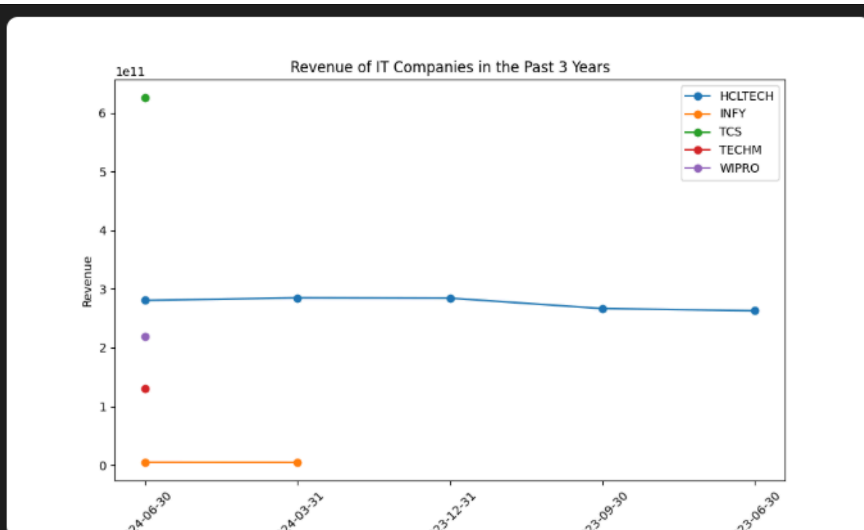
Show me the revenue of IT companies in the past 3 years.

Response:

Revenue of IT companies in the past 3 years:

	date	symbol	revenue
0	2024-06-30	HCLTECH	280570000000
1	2024-06-30	INFY	4714000000
2	2024-06-30	TCS	626130000000
3	2024-06-30	TECHM	130055000000
4	2024-06-30	WIPRO	219638000000
5	2024-03-31	HCLTECH	284990000000
6	2023-12-31	HCLTECH	284460000000
7	2023-09-30	HCLTECH	266720000000
8	2023-06-30	HCLTECH	262960000000
9	2024-03-31	INFY	4565000000





Further suggestions:

You can ask for the total revenue of specific IT companies for each year over the past three years. Additionally, consider requesting a comparison of revenue growth rates among these companies during that period.

Prompt: *What kind of data do you have*

What kind of data do you have

Response:
The database contains financial data for various companies, organized into four main tables.

1. **Stock Prices**: This table includes daily stock price information such as opening, closing, high, and low prices, along with trading volume for each company identified by its symbol.
2. **Income Statement**: This table provides detailed financial performance metrics, including revenue, expenses, net income, and earnings per share (EPS) for each company, along with relevant dates and currency information.
3. **Balance Sheet Statement:** This table captures the financial position of companies, detailing assets, liabilities, and equity components, including cash, receivables, and total liabilities.
4. **Cash Flow Statement:** This table outlines cash inflows and outflows from operating, investing, and financing activities, including net cash provided by operations and capital expenditures.

Each table is linked by the company symbol, allowing for comprehensive financial analysis.

Prompt (Out of Context): *Hello*

Hello

Response:
The request is out of context.

Further suggestions:

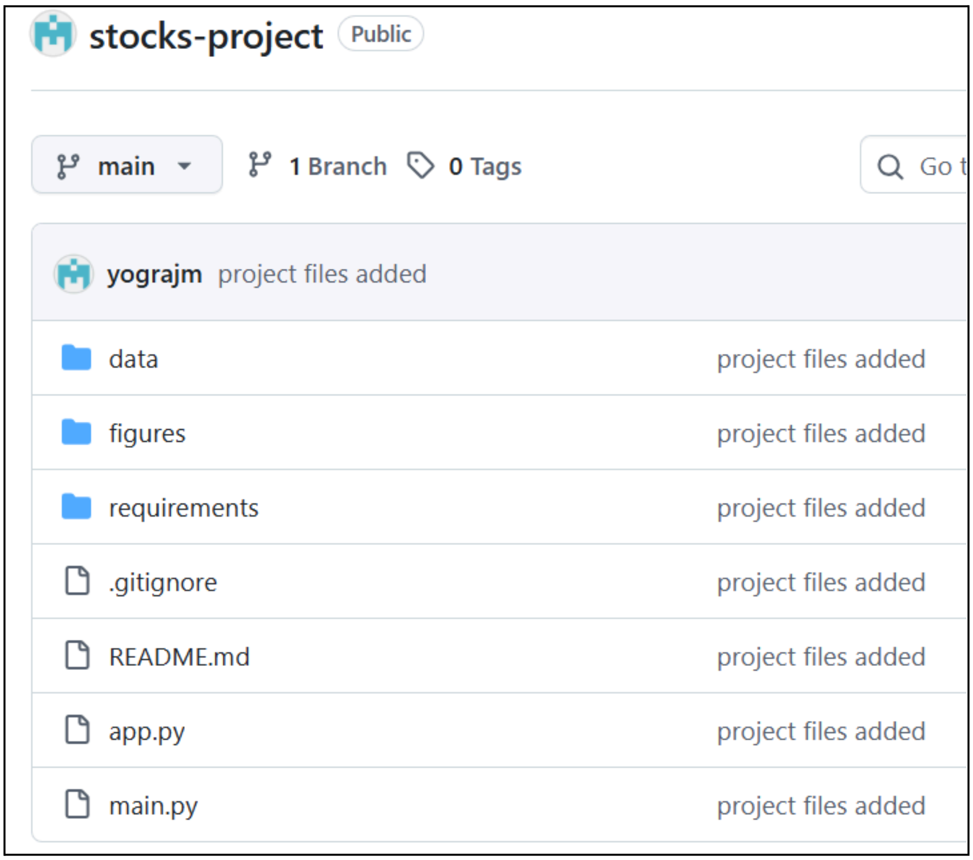
1. Ask for the historical stock prices of a specific company over the last year.
2. Inquire about the latest income statement details for a particular company, including revenue and net income.
3. Request a comparison of total assets and liabilities for a company over the last few years.
4. Seek insights on cash flow trends, such as net cash provided by operating activities for a specific company.

20. Once done, close the application by pressing Ctrl + C in the terminal.

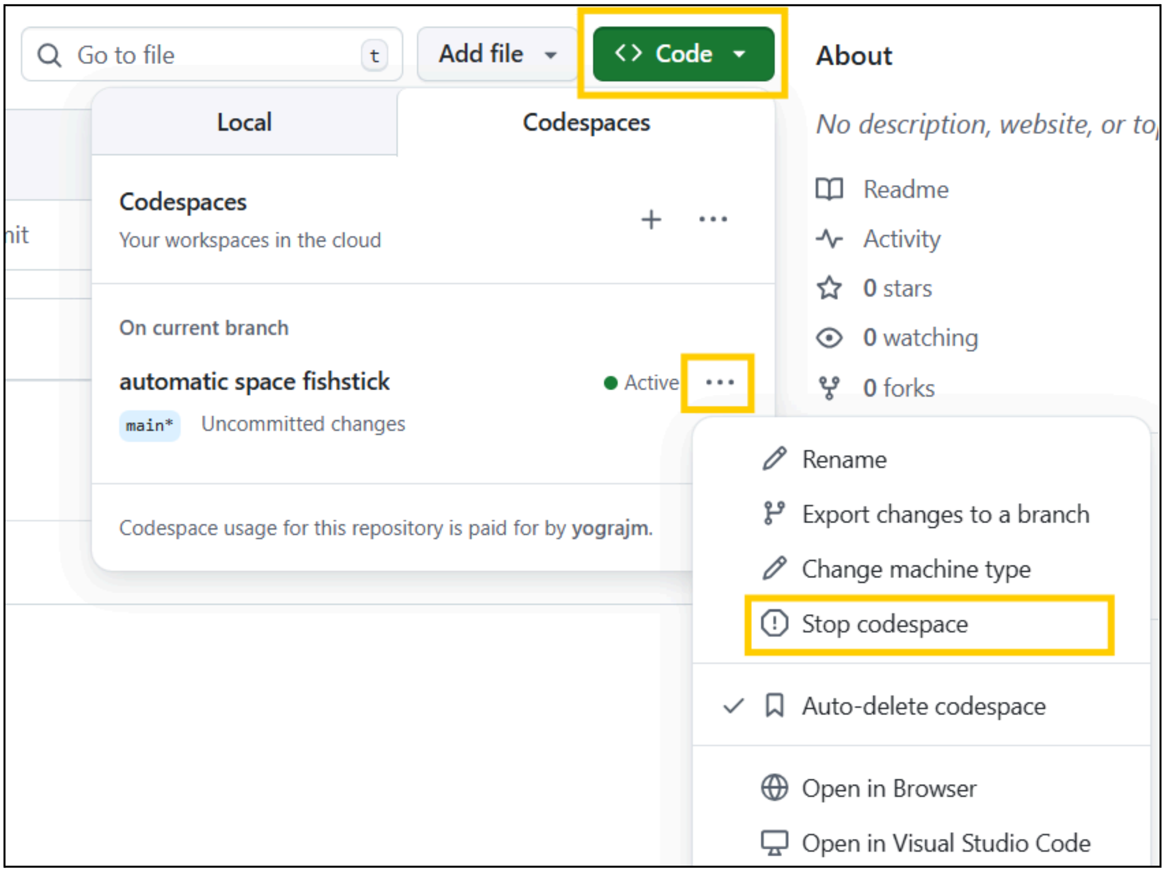
21. Run the below commands to push the files present in your GitHub Codespace to your remote GitHub repository.

```
git add .
git commit -m "project files added"
git push
```

22. Check your GitHub repository, the files should be visible.



23. At last, you can stop & delete the Codespace.


Go to file
t
Add file
Code
About
Local
Codespaces
No description, website, or to
Codespaces
Your workspaces in the cloud
Readme
Activity
0 stars
0 watching
0 forks
On current branch
automatic space fishstick
Active
main*
Uncommitted changes
Rename
Export changes to a branch
Change machine type
Stop codespace
Auto-delete codespace
Open in Browser
Open in Visual Studio Code
Codespace usage for this repository is paid for by yograjm.

~~~END~~~